

These positions are funnelled into the squad leader's maps from intelligence sources like the Predator unmanned aerial vehicle. The computer also ties into the soldier's communications system, from which he can read and send real-time text messages via a voice interface to other squad members and view still images from his rifle camera. As a result, Land Warrior-equipped soldiers are able to cover more ground with fewer men, see farther, shoot with greater accuracy and communicate more covertly.

DARPA has also developed lightweight, low-cost electronic yarns that can be woven into a network of sensors, actuators, logic algorithms and power sources, and can be twisted and crumpled as any other clothing. These smart uniforms are an example of how the newest technologies are improving the efficiency, safety and power of the oldest weapons platform of all—the foot soldier. It can also relay GPS information, and the design includes Interceptor body armour, designed to stop small fragmentation munitions.

Transforming healthcare

Wearable technology will enable surgeons to monitor a patient's vital parameters and react to variations at the same time—without ever having to lose sight of the procedure or the patient. Hence, doctors will have all the key information and facts, hands-free, in split seconds, when they need it most, say in the operation theatre.

Google Glass is a wearable technology device that has a tiny prism on the right side to display information by means of a Wi-Fi or Bluetooth connection to the MyGlass app on Android and even iOS devices. Researchers from Philips, in collaboration with Accenture Technology, have been experimenting to discover the possible uses of Google Glass in healthcare since it allows doctors to 'virtually' be in two places simultaneously, thus making patient care so much more efficient. Imagine a surgeon or nurse attending to a patient and carrying out a procedure that needs both hands. Suddenly



Rewalk exoskeleton: Wearable robotic legs
(Source: <http://www.glorysurgery.com/>)

an alarm is sounded in the next room, but the doctor does not have to immediately abandon the current procedure. He communicates with the patient monitoring data system that has sent the alert and then takes a call if what's happening in the adjoining room can wait or needs immediate action.

The other area where wearable computing is transforming healthcare is sensors-based tools. These are generally made up of gold conductive lines and ultra-thin sheets of silicon, and can transmit electric signals to the skin. Sensor-based tools enable doctors to feel the pulse rate, pressure and other vital signs more accurately, resulting in a more precise surgery. This novel tool is based on three basic ideas:

1. Using a special process, thinner flexible silicon can be developed which is more elastic
2. Semiconductor devices and arrays of the micro materials can be printed instead of being forged
3. Good flexibility can be provided by stretchable interconnects that connect arrays about 200 micrometres apart.

This results in 'flexible electronics' and is behind monitoring devices that include LED and photo sensors as well as components. These sensors are so powerful that they can identify if there is a disease or not, and the intensity,



The MindMesh, an EEG-based 'thinking cap'
(Source: <http://www.interaction-design.org/>)

from samples of blood, urine, etc itself, without the requirement of any lab equipment. These sensors can be put on catheters to sense heart conditions in real-time basis, sewn into gloves so that surgeons can use their fingers as sensing instruments during surgery and even be placed on body parts, such as arms, like a bandage. This new technology could yield surgical robots that can network with their environs via touch.

Hence we are moving towards incorporation of multifunctional silicon semiconductor device technologies in the shape of soft, 3D, form-fitting skins, apt for wearing not only on the fingertips but also other body parts as well. For example, a device could encase the complete 3D surface of the heart to facilitate different sensing and stimulating functions, thus allowing advanced surgical and diagnostic possibilities.

The powerful powered exoskeleton

In the movie *Batman: The Dark Knight Returns - Part 2*, Batman puts on a powered exoskeleton for his combat with Superman. This enables him to lift the Batmobile easily with one hand and fight on equal terms with Superman. A powered exoskeleton is a mobile machine and also goes by the name of powered armour, exoframe,